

SmartInv

Multimodal Smart Contract Invariant Inference

A★ PAPER EVALUATION

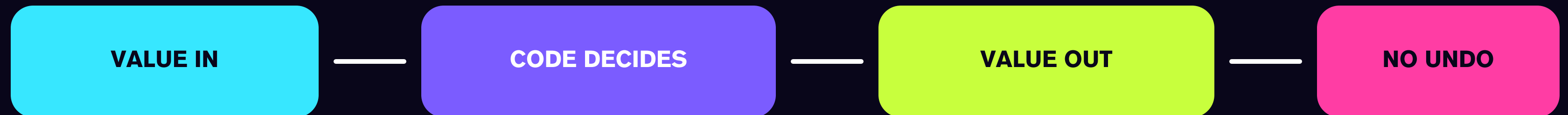
Mid-semester project review
Rishi · 2024ucp1566



Why one tiny bug can move billions

A smart contract is like a vending machine: insert value, rules run, output appears. But once deployed, the machine is public, automatic—and painfully hard to rewind.

\$3.8B
stolen in 2022



Same codebase. Two very different enemies.

IMPLEMENTATION BUG

The machine is built wrong

Overflow, reentrancy, access control. Think: a loose wire or jammed coin slot. Static analyzers are good here.

FUNCTIONAL LOGIC FLAW

The machine follows the wrong rule

The code runs exactly as written—but violates business intent. Think: dispensing two products for one payment.

\$8.2M vanished through a rule nobody checked

The attacker supplied fake deposit data. The contract accepted it because a crucial relationship was never enforced. Syntax scanners saw valid code; the business rule was missing.

TRADITIONAL TOOLS: PASSED

Valid syntax $\neq$ valid intent

An invariant is a promise that must always stay true.

01

Elevator

Doors stay closed while moving.

02

Bank account

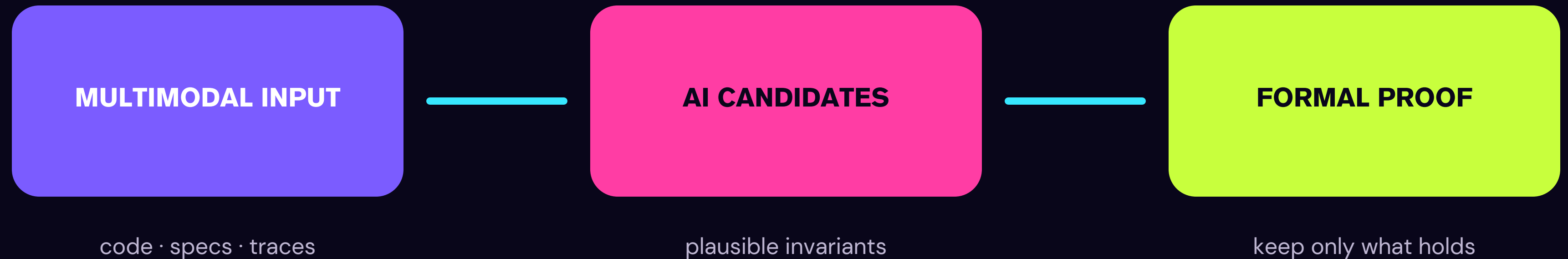
Balance never drops below allowed limits.

03

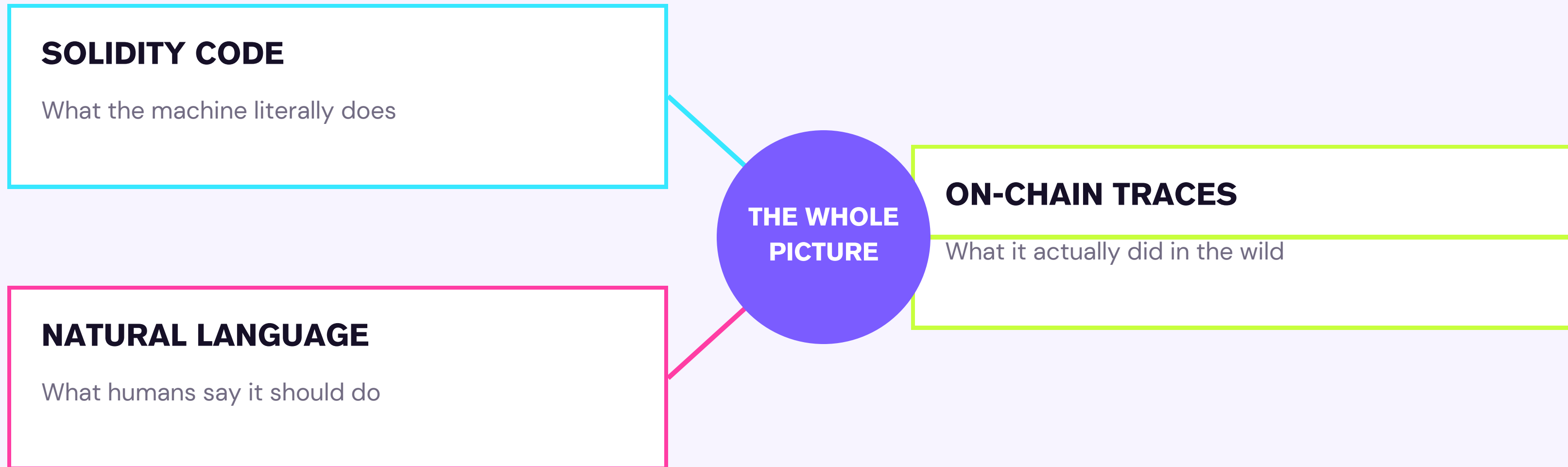
Token vault

Shares always match the assets inside.

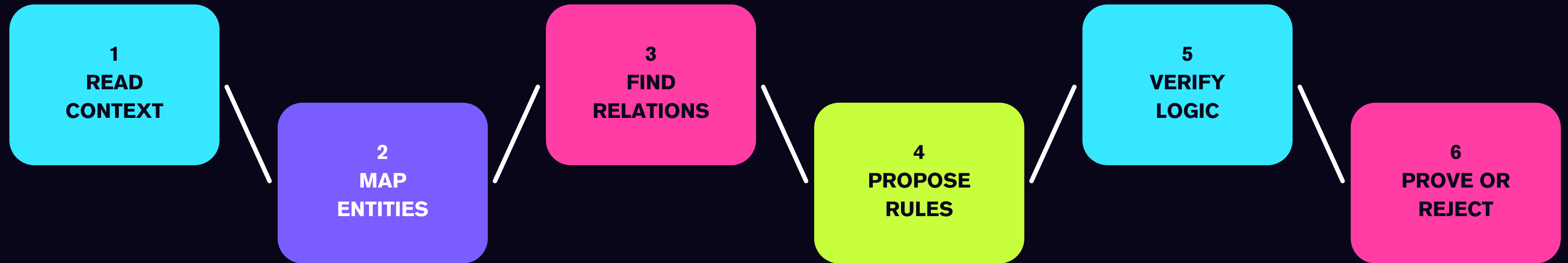
AI guesses the rule. Math checks the homework.



One contract. Three ways to understand it.

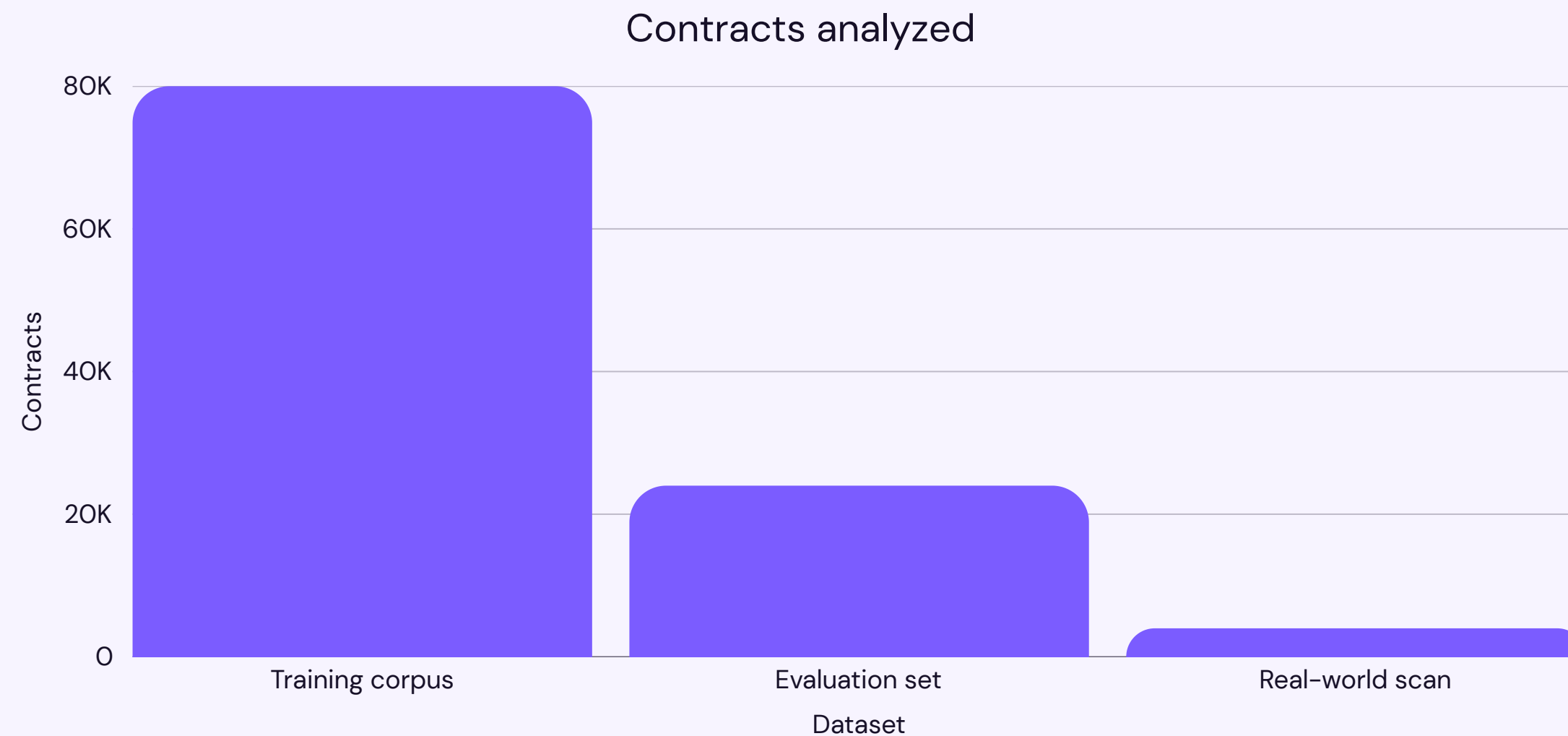


Six small reasoning steps beat one giant guess.



Structure is the guardrail against hallucination.

Tested at ecosystem scale—not on toy examples.

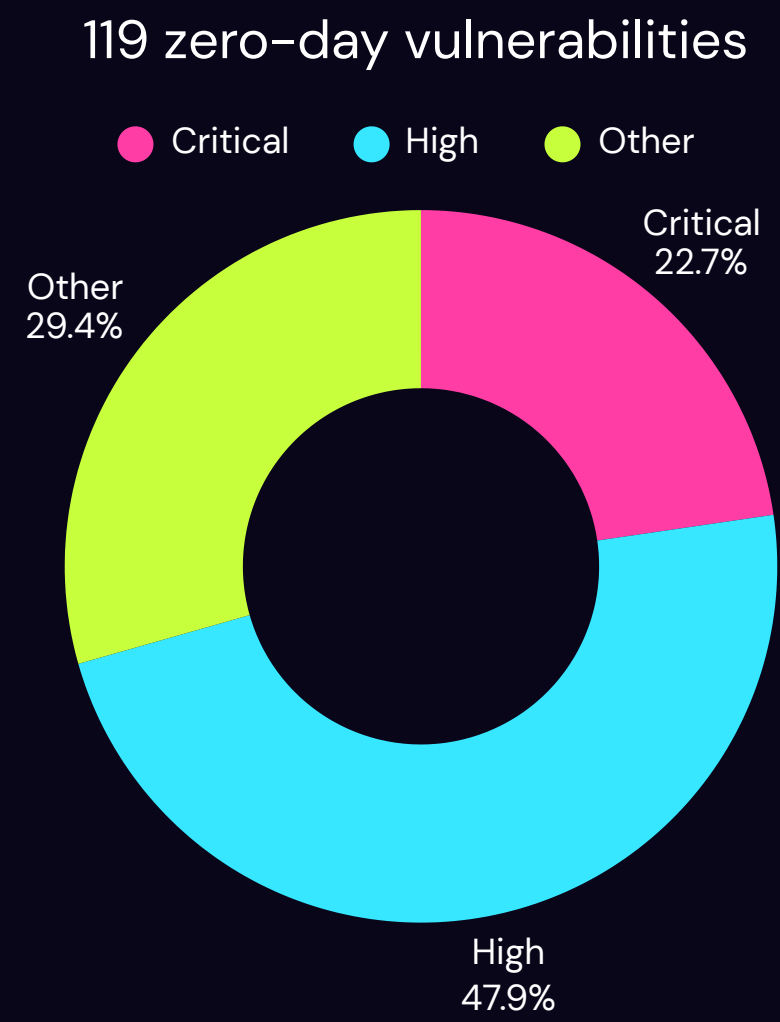


80,000+

real Ethereum contracts
formed the learning base.

Scale matters: rare logic patterns
only appear when the dataset is
broad.

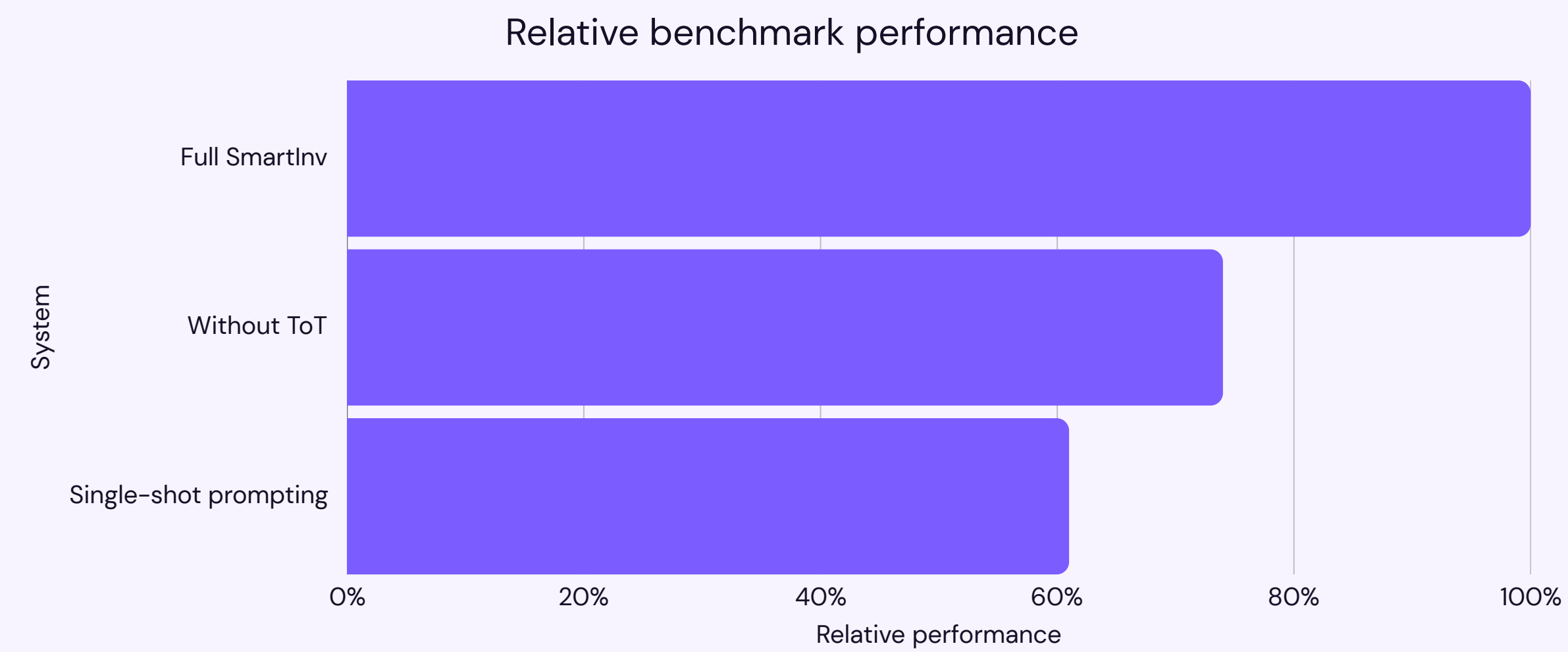
119 bugs nobody knew existed.



119
ZERO-DAYS

Fresh mainnet findings turn SmartInv from a clever paper into a practical security system.

Remove Tier of Thought—and the system loses its edge.



ToT is the hero

It converts fuzzy model intuition into a repeatable checklist—raising precision and making errors easier to inspect.

From paper to a working local pipeline.

01 ENVIRONMENT

Reproduced the stack locally on Windows.

02 BUG FIXES

Patched path, dependency and shell issues.

03 DATASET

Generated contract samples and metadata.

04 PIPELINE

Validated end-to-end inference stages.

SmartInv sees what syntax misses.

Multimodal context finds the intent. Tier of Thought structures the reasoning. Formal verification keeps every claim honest.

AI INTUITION × MATHEMATICAL PROOF

Questions? Let's stress-test the idea.

